

Tendling: Designing An Embodied AI Pillow for Everyday Emotion Regulation

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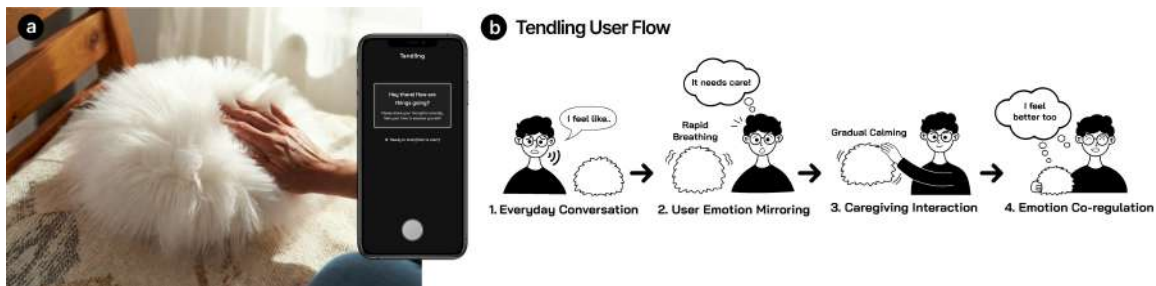


Figure 1: (a) Tendling: Embodied AI pillow for everyday emotion regulation (b) Tendling user flow.

Abstract

This paper presents Tendling, a palm-sized breathing pillow that explores embodied empathy in human-AI interaction through emotion co-regulation. While existing tangible interfaces for emotion regulation primarily serve as biofeedback tools, Tendling explores physical forms of relationship in which an AI agent somatically interacts the user's emotional state through respiratory patterns. Our prototype utilizes multimodal input consisting of use conversation and touch input, where the user's gentle touch calms the AI device's breathing while violent handling intensifies tension. Based on Self-Determination Theory and recent findings on prosocial behavior toward AI agents, we explore how we can implement embodied AI to support mutual emotion regulation through caregiving interactions. We offer a detailed critical-reflective account of our process of making Tendling into a system for emotional empathy, and discuss design challenges in translating emotional states into tactile responses. We discuss how these insights can inform future embodied AI designs for emotional support.

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CCS Concepts

• Human-centered computing; • Human computer interaction (HCI); • Interaction devices;

Keywords

AI Embodiment, Emotion Regulation, Prosocial Behavior, Tangible Interaction

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1 INTRODUCTION and Related Work

This paper presents Tendling (Figure 1), a small breathing pillow that explores embodied empathy in human-AI interaction for everyday emotion regulation. Emotion regulation refers to processes by which individuals influence which emotions they have and how they experience and express them [12]. Using research through design (RtD) methodology [25, 32], we investigate what embodied AI interaction design should consider when supporting emotion regulation through prosocial behaviors, voluntary actions intended to benefit others, including helping, sharing, and caregiving [29]. We present a critical account of our design process and challenges

encountered through iterative prototyping of an AI embodiment system for everyday emotion regulation.

Our project is initiated to address two converging needs in everyday emotion regulation technology. First, while everyday technologies for emotion regulation (e.g., mental health apps, meditation services) have proliferated to support emotional well-being, young adults face elevated mental health risks. A 2024 survey of 95,000 students (age 18-25) across 373 universities reported 38% experiencing depression, with suicide rates increasing from 11.6 to 18.1 per 100,000 [8]. As young adulthood is the period for forming lifelong regulation habits [1], developing emotion regulation technologies for young adults remains an important task in HCI. Second, existing emotion regulation technologies frequently rely on digital notifications, logging, or guided exercises asking users to articulate emotions, follow instructions, or reframe thoughts through a screen [2]. [15] argues such approaches prioritize efficiency over emotional resonance, distancing users from physiological states. Beyond this, interventions that require conscious reflection impose emotional burden [17], and clinical framing can hinder routine integration [6, 27].

To address this, recent HCI research suggests two complementary paths: One comes from somaesthetic interaction design, which treats emotion regulation as fundamentally embodied, requiring engagement with bodily sensations rather than only cognitive strategies (e.g., deliberate self-reflection) [15, 24]. Systems such as Soma Mat and Breathing Light guide attention to breathing through tangible rhythmic feedback [24], but often remain in biological self-monitoring, preserving the feeling that technology is a clinical or therapeutic tool. The second path comes from HCI research on prosocial behavior and well-being. Psychological work shows that prosocial behaviors enhance well-being through experiencing pleasure and meaning in life [22, 29]. [29] suggests that autonomous helping supports positive affect and meaning by satisfying needs for autonomy, competence, and relatedness. Self-Determination Theory frames these needs as universal, with their satisfaction being a key mechanism through which everyday behaviors influence well-being [22]. [11] shows that even brief prosocial micro acts increase happiness by strengthening perceived competence and social connection. Interestingly, recent HCI findings extend these benefits beyond human-to-human contexts. [31] demonstrates that prosocial actions toward AI agents reduce loneliness and increase positive affect while AI fulfills competence and autonomy needs. Human-robot interaction works also show virtual pets and social robots elicit caregiving and routines supporting mental health [7, 23].

Taken together, these literatures suggest an underexplored design opportunity for embodied AI that invites physical caring interaction as an indirect route to emotion regulation [29, 31]. Unlike systems leaning on the analytical interface [20, 30], Tendling positions everyday emotion regulation technology as a vulnerable entity that requires care rather than a high-technology tool. Silent throughout, Tendling communicates only through respiratory patterns reflecting users' emotional arousal (e.g., stress, calm). Building on prior empirical evidence [31] that helping AI agents reduces loneliness and enhances positive affect for well-being, we explore a new embodied AI interaction to satisfy users' needs for competence (e.g., seeing the impact of one's actions) and autonomy (e.g., experiencing voluntary choice) for emotion regulation. By making

AI well-being visible and responsive, Tendling reframes emotion regulation experiences as a byproduct of caregiving rather than explicit therapy.

2 DEVELOPMENT OF TENDLING

Tendling is a small interactive pillow designed to suggest embodied empathy through caregiving in human-AI interaction (Figure 1). After conversational exchanges where an AI agent interprets emotional state through utterances and voice analysis, that interpretation becomes tactile breathing. Gentle stroking and patting slow the breathing, while rough handling increases agitation. Tendling synthesizes conversational intelligence with abstract physical embodiment, producing an entity that is neither animal nor object yet can express distress and respond to care. Drawing on caregiving behaviors linked to soothing and emotion regulation [26, 29, 31], we posit that responsive AI well-being can satisfy competence and autonomy needs in ways that support emotion regulation without explicit clinical framing. In line with prior concerns about clinical questioning and interview style agents that create burden or social distance [5, 21], we aim to understand user emotion in natural conversations with Tendling. Tendling connects an AI conversation stage to an embodiment stage expressing inferred emotional arousal as breathing, then links the user reaction and treatment to breathing modulation. Through iterative prototyping, our development required integrating three components: (1) extracting emotion from natural conversation, (2) implementing living-like breathing interaction, and (3) integrating system and materiality into an embodied AI object that responds legibly to user actions.

2.1 Research-Through-Design Methodology

Following RtD approach [25, 32], we developed Tendling via iterative prototyping to address unanticipated empirical gaps between theoretical strategies and technical implementation. We incorporated pilot testing (October 2 to December 4, 2025) with undergraduate students (N=3; all females; ages 22-24) recruited from the authors' university. Participants met twice weekly in a focus group format, experienced at least four prototype versions (12-15 sessions in total), and provided feedback. This aligns with small-scale iterative testing to identify usability issues before resource-intensive studies [16]. Although the sample cannot represent diverse populations, extended engagement enabled participants to compare iterations and articulate nuanced issues that brief usability tests may not reveal [18].

2.2 Implementation and Design Iteration

2.2.1 Stage 1: Extracting Emotion from Natural Conversation. To externalize and mirror user emotion through breathing interaction, the first design challenge was extracting emotion from everyday conversation without asking users to explicitly label feelings through clinical and analytical prompts. We needed to balance low-burden interaction with coherent emotional understanding. We implemented a multimodal emotion detection pipeline using OpenAI API (GPT-4) with JavaScript, processing audio and text transcription by extracting volume, speech rate, pause duration, and vocal variation via Web Audio API (Figure 2).



Figure 2: (a) initial prototype for emotion detection (b) Tendling AI conversation flow (c) Responsive Breathing Interaction.

Table 1: A multi-modal system to extract emotion from AI conversation

Category	Priority	Description
Explicit Emotion Statements	highest priority	Pattern detection for phrases such as “I feel” or “I’m feeling” followed by emotion words. These override other indicators.
Lexical Analysis	40% weight	Keyword matching from transcriptions against our typology with intensity scoring based on frequency and context. For example, “exhausted” receives higher intensity than “tired” when detecting fatigue-related emotions.
Audio Prosody Analysis	20% weight	Volume, speech rate, pause duration, and pitch variation provide embodied emotional cues. Increased volume with faster speech indicates heightened arousal, such as anxiety.
Contextual Analysis	30% weight	Examination of conversation topics (work stress, relationship issues), response length patterns, and overall sentiment trends reveals indirect emotional states.

First, we began with Emotion Typology [10] to measure both positive and negative emotions, but early testing revealed mismatches between theoretical definitions and how people naturally express emotions in conversation. Many emotions defined in the framework proved difficult for LLMs to detect automatically in everyday dialogue, as definitions did not align with the language people actually used when discussing their episodes and emotional states. We developed a hybrid approach by analyzing common expressions in user dialogues and grouping them into three-tier typologies grounded in empirically observed phrasing. This aligns with [4] grounded emotion recognition, emphasizing the need to match computational models to naturally occurring language patterns rather than theoretical constructs alone. Our typology contained primary emotions (joy, sadness, anger, fear, surprise, disgust [9]), secondary emotions (love, shame, pride, envy, gratitude, hope, despair) reflecting complex social-evaluative states [28], and tertiary emotions (calm, excitement, boredom, confusion, relief, loneliness) capturing everyday affective experiences often missing from basic emotion frameworks [3]. Each category included keyword sets from how participants naturally spoke, improving automatic detection while preserving coherent structure.

Second, we encountered tension between contextual understanding and conversational naturalness. Direct emotion elicitation creates social distance and reduces self-disclosure [21], while casual conversation supports spontaneous expression [14]. P3 noted early versions asking, “What emotions are you experiencing?” felt “like talking to a therapist app.” We developed a multimodal weighting system (Table 1) inferring emotion through converging lexical, prosodic, and contextual signals. This enabled reliable detection without clinical prompts. When P3 said “Work has been really

piling up” with increased volume and faster speech, the system detected anxiety through converging cues.

Third, we calibrated conversational turns needed to gather sufficient context without causing fatigue. Single-turn exchanges yielded insufficient data, with participants rarely sharing genuine feelings. P1 explained, “It feels too abrupt to talk about how I’m really feeling right away.” Five exchanges caused fatigue by the fourth turn, “I just want to get to the breathing part already (P1).” We set the three-turn conversation, providing enough context while maintaining engagement. Participants appreciated this as “like checking in with a friend (P2).”

2.2.2 Stage 2: Implementing Breathing Interaction That Mirrors User Emotion. Given that human-robot interaction study demonstrates that lifelike movement patterns enhance perceptions of aliveness and foster emotional connection [13], designing breathing interaction was needed to feel alive to position Tendling as a vulnerable entity worthy of care. We implemented an inflate-deflate mechanism using Arduino Uno R4 with LM298N dual motor driver controlling two air pumps (Figure 3). Pumps connected through a long tube to a balloon inside Tendling’s fabric body. We mapped detected arousal values ($range = 0-255$) to breathing rhythms using PWM (Pulse Width Modulation) control. Early prototypes used equal inflation and deflation durations, but participants experienced patterns as artificial and awkward, “It feels more like a ball inflating and deflating than actual breathing (P1).” We introduced transition delays so inhalation (60% duration) lasted longer than exhalation (40% duration), approximating natural respiratory dynamics (Jerath et al., 2015). We also calibrated gradual shifts between breathing states in three breathing zones of Slow (4-6s cycles), Moderate (2.5-4s), and Fast (1.5-2.5s). After these modifications, P2 described

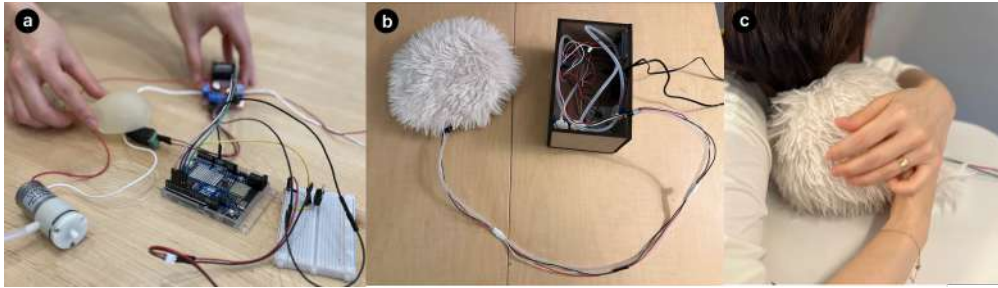


Figure 3: (a) Development of Breathing Interaction (b) Integrating System with Materials (c) Final Prototype.

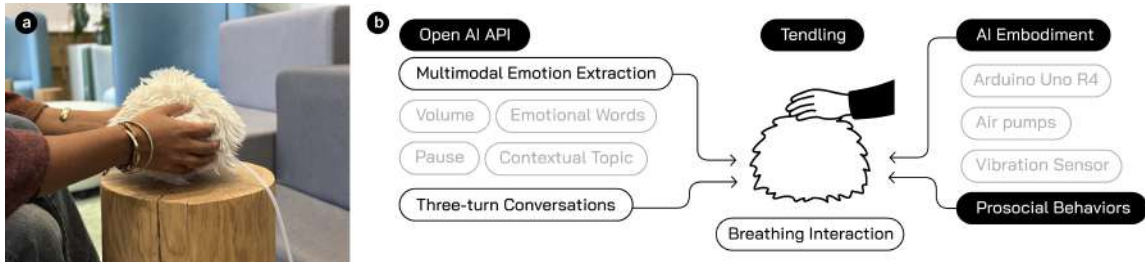


Figure 4: (a) Participants' voluntary caregiving interaction in the pilot test (b) Tendling AI Embodiment Mechanism.

breathing as “It feels more alive when I place my hands on it than just watching it. It’s like feeling a heartbeat, or like a tiny puppy curled up breathing.”

2.2.3 Stage 3: Integrating System and Material Towards AI Embodiment. The third challenge was integrating conversation-based inference, breathing embodiment, and touch responsiveness into a coherent experience of vulnerability and care (Figure 4). We selected a flexible, fluffy cotton blend to support cradling and gentle touch. Size strongly influenced perceived neediness and breathing readability. Through testing, we arrived at a 22 cm diameter fitting in both hands. P2 said larger prototypes were “good for hugging, but don’t look like something needing care,” while P1 found smaller versions insufficient because “I cannot see breathing in and out.” This size reinforced Tendling as vulnerable yet large enough to make breathing perceptible. To detect the user’s physical treatment, we used a vibration sensor. Sensor placement required balancing sensitivity against environmental noise. For example, near-surface placement reacted to incidental movements, “I’m just adjusting how I’m holding it, and suddenly it thinks I’m trying to calm it down (P3).” When embedded too deeply, gentle touches were missed, “I’m petting it really gently, but nothing’s happening (P1).” We decided to place the sensor approximately 3 mm beneath the fabric and adjusted the sensor to reduce environmental vibration while remaining sensitive to intentional gestures.

Lastly, we developed a two-phase control structure to coordinate conversation-based emotion detection with real-time touch responsiveness. P1 commented, “I can’t tell if it’s reacting to me or if something’s wrong.” We implemented two phase interaction: Phase 1 establishes an initial breathing pattern based on inferred arousal for 25 seconds, allowing users to observe Tendling’s state.

Phase 2 enables touch-responsive behavior, where caregiving gestures gradually modulate breathing rather than instantly overriding it, thereby creating a mutual influence. P1 reflected, “I can see it’s stressed at first, and then as I pet it, it gradually calms down. It feels like I’m actually helping.” Over repeated sessions, participants took spontaneous caregiving gestures without instruction. P1 cradled Tendling, and P2 showed consistent stroking patterns. These behaviors suggest that embodied caregiving interactions may help enjoy and integrate emotion regulation into daily routines through voluntary soothing acts.

3 DISCUSSION AND CONCLUSION

This paper offers empirical insights into integrating conversational intelligence, physical implementation, and lifelike responsiveness for embodied AI caregiving interactions. Building on [31], we explored design challenges to invite users in AI embodiment interaction to support everyday emotion regulation. First, we contribute a conversationally grounded emotion typology and multimodal extraction approach, bridging emotion theory with everyday conversation. By combining explicit statements, lexical cues, prosody, and context, we inferred arousal while avoiding direct emotion elicitation perceived as clinical. This extends conversational agent design by showing how natural dialogue reduces burden and preserves authenticity in everyday use [14, 21, 27]. Second, we adjusted breathing interaction and materials to improve realism for the living artifact, with participants describing touch as “like feeling a heartbeat (P2).” This aligns with somaesthetic interaction design by emphasizing bodily perception as a pathway to engagement [15, 24] in shifting the user role from self-monitoring to soothing another. Rather than treating AI as a high-technological tool for emotion regulation, Tendling frames it as a vulnerable artifact that

synchronizes with user emotion, aligning with well-being mechanisms in Self-Determination Theory [22, 29] and relational artifact perspectives inviting care [19].

However, the homogeneity of our small sample limits the generalizability of our findings across emotional expressions, caregiving styles, and cultural communication patterns. Variations in caring interaction style or technological familiarity may shape user experience in various ways that this study could not capture. Future work should investigate expanded design considerations for more diverse populations, including varied cultural and gender contexts, and validate the effectiveness of emotion regulation through Tendling in everyday living contexts.

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